The documentation and process conversion measures necessary to comply with this revision shall be completed by 7 September 2026.

METRIC

MIL-PRF-19500/741B
w/AMENDMENT 4
7 June 2026
SUPERSEDING
MIL-PRF-19500/741B
w/AMENDMENT 3
13 November 2025

PERFORMANCE SPECIFICATION SHEET

# TRANSISTOR, DIE, FIELD EFFECT, RADIATION HARDENED (TOTAL DOSE AND SINGLE EVENT EFFECTS) N-CHANNEL AND P-CHANNEL, SILICON, VARIOUS TYPES, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

## 1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel and P-channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event effects (SEE)), power transistor die. Two levels of product assurance (JANHC and JANKC) are provided for each device type as specified in MIL-PRF-19500. Provisions for radiation hardness assurance (RHA) to four radiation levels ("G", "H", "R" and "F") are provided for JANTXV and JANS product assurance levels.

1.2 Physical dimensions. See figures 1 through 15 herein.

1.3 Maximum ratings. See the applicable performance specification sheet from table I herein.

1.4 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.4.1 JAN certification mark and quality level for un-encapsulated devices (die). The quality level designators for un-encapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.4.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "R", "F", "G", and "H". Since the RHA levels vary on different slash sheets, see 6.5 for available PINs.

Comments, suggestions, or questions on this document should be addressed to DLA Weapons Support, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A
FSC 5961
DISTRIBUTION STATEMENT A. Approved for public release. Distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

1.4.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.4.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.4.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "7467", "7468", "7469", "7470" "7471", "7472", "7473", "7474", "7475", "7476", "7477", "7478", "7479", "7480", "7481", "7482", "7483", "7484", "7485", "7486", “7488", "7489", "7490", "7491" "7492", "7493", "7494", "7495", "7496", "7497", "7498", "7500", "7501", "7502", "7503", "7506", "7519", "7520", "7521", "7522", "7523", "7524", "7525", "7526", "7545", "7546", "7547", "7548", "7549", "7550", “7555”, “7561”, "7616", "7626". and "7656".

1.4.5 Suffix symbols. Suffix symbols are not applicable for this specification sheet.

1.4.6 Die identifiers for un-encapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifier that is applicable for this specification sheet is "A" (see figures 1 through 15 and 6.5).

* 2. APPLICABLE PUBLICATIONS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

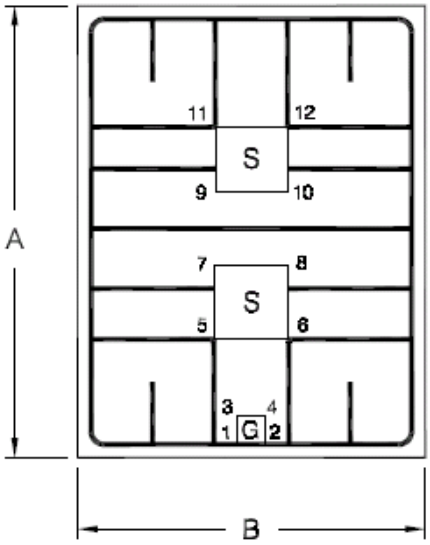


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7467, 2N7478, 2N7468, 2N7470**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 8634.0 | 339.9 | ±200 µm (7.87 mils) |
| B | 6602.0 | 259.9 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 541.00 | 0 | 21.299 | 0 | |
| 3 | 0 | 538.30 | 0 | 21.193 | |
| 4 | 541.0 | 538.30 | 21.299 | 21.193 | |
| 5 | -424.1 | 2005.1 | -16.7 | 78.941 | |
| 6 | 969.0 | 2005.1 | 38.15 | 78.941 | |
| 7 | -424.1 | 3397.5 | -16.7 | 133.76 | |
| 8 | 969.0 | 3397.5 | 38.15 | 133.76 | |
| 9 | -424.1 | 4814.1 | -16.7 | 189.53 | |
| 10 | 965.1 | 4814.1 | 37.996 | 189.53 | |
| 11 | -424.1 | 6033.1 | -16.7 | 237.52 | |
| 12 | 965.1 | 6033.1 | 37.996 | 237.52 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.6 µm (0.22 mils) thick.
3. Back metal: Chromium, nickel, Vanadium-silver, 0.1, 0.2, 0.5 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 305 µm (12 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 12 refer to adjacent gate(G)/source(S) pad corners. Key 2 through 12 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 1. JANHC and JANKC A-version die dimensions, 2N7467, 2N7478, 2N7468, and 2N7470.

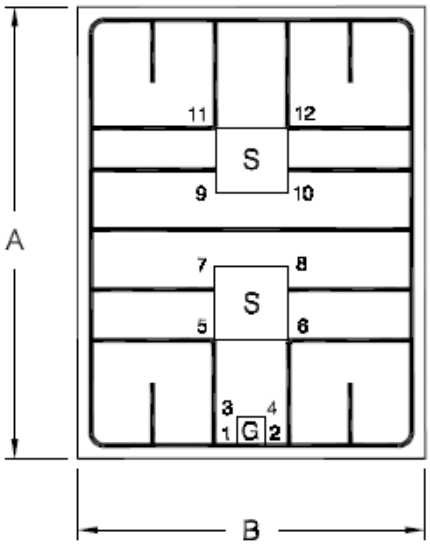


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7469, 2N7471, 2N7472, 2N7475, 2N7473, 2N7476, 2N7474, 2N7477**

| Ltr. | Dimensions | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | Min | Max | Min | Max | |
| A | 9124.0 | | 359.21 | | ±200 µm (7.87 mils) |
| B | 6582.0 | | 259.13 | | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 541.0 | 0 | 21.299 | 0 | |
| 3 | 0 | 538.3 | 0 | 21.193 | |
| 4 | 541.0 | 538.3 | 21.299 | 21.193 | |
| 5 | -424.1 | 2199.6 | -16.7 | 86.598 | |
| 6 | 969.0 | 2199.6 | 38.150 | 86.598 | |
| 7 | -424.1 | 3592.0 | -16.7 | 141.42 | |
| 8 | 969.0 | 3592.0 | 38.150 | 141.42 | |
| 9 | -424.1 | 5013.9 | -16.7 | 197.40 | |
| 10 | 965.1 | 5013.9 | 37.996 | 197.40 | |
| 11 | -424.1 | 6232.9 | -16.7 | 245.39 | |
| 12 | 965.1 | 6232.9 | 37.996 | 245.39 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.5 µm (0.22 mils) thick.
3. Back metal: Ti, NiV, Ag, 0.1, 0.45, 0.6 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 290 µm (11.4 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 12 refer to adjacent gate (G)/source (S) pad corners. Key 2 through 12 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 2. JANHC and JANKC A-version die dimensions, 2N7469, 2N7471, 2N7472, 2N7475, 2N7473, 2N7476, 2N7474, 2N7477.

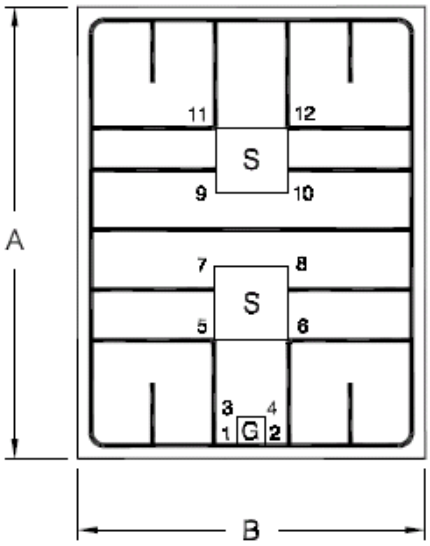


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7549, 2N7550**

| Ltr. | Dimensions | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | Min | Max | Min | Max | |
| A | 9124.0 | | 359.21 | | ±200 µm (7.87 mils) |
| B | 6582.0 | | 259.13 | | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 541.0 | 0 | 21.299 | 0 | |
| 3 | 0 | 538.3 | 0 | 21.193 | |
| 4 | 541.0 | 538.3 | 21.299 | 21.193 | |
| 5 | -424.1 | 2199.6 | -16.7 | 86.598 | |
| 6 | 969.0 | 2199.6 | 38.150 | 86.598 | |
| 7 | -424.1 | 3592.0 | -16.7 | 141.42 | |
| 8 | 969.0 | 3592.0 | 38.150 | 141.42 | |
| 9 | -424.1 | 5013.9 | -16.7 | 197.40 | |
| 10 | 965.1 | 5013.9 | 37.996 | 197.40 | |
| 11 | -424.1 | 6232.9 | -16.7 | 245.39 | |
| 12 | 965.1 | 6232.9 | 37.996 | 245.39 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.5 µm (0.22 mils) thick.
3. Back metal: Al, Ti, NiV, Ag, 0.2, 0.2, 0.45, 0.6 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 240 µm (9.4 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 12 refer to adjacent gate (G)/source (S) pad corners. Key 2 through 12 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 3. JANHC and JANKC A-version die dimensions, 2N7549, 2N7550.

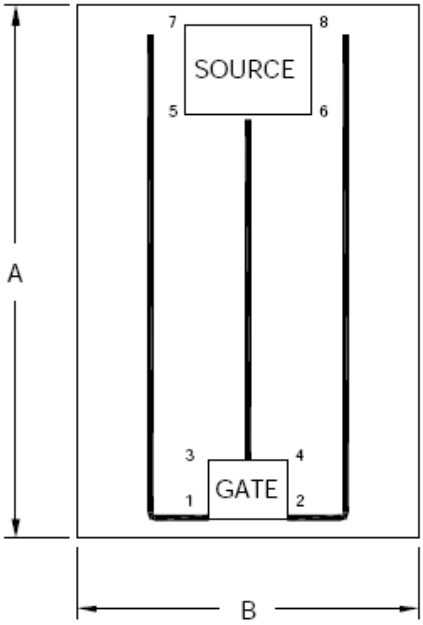


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7479, 2N7482, 2N7491, 2N7494, 2N7480, 2N7483, 2N7492, 2N7495**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4604.3 | 181.3 | ±200 µm (7.87 mils) |
| B | 2946.4 | 116.0 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 509.9 | 0 | 20.075 | |
| 4 | 685.9 | 509.9 | 27.004 | 20.075 | |
| 5 | -206.1 | 3474.3 | -8.114 | 136.78 | |
| 6 | 871.9 | 3474.3 | 34.327 | 136.78 | |
| 7 | -206.1 | 4235.4 | -8.114 | 166.75 | |
| 8 | 871.9 | 4235.4 | 34.327 | 166.75 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.6 µm (0.22 mils) thick.
3. Back metal: Chromium, nickel, Vanadium-silver, 0.1, 0.2, 0.5 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 305 µm (12 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 4. JANHC and JANKC A-version die dimensions, 2N7479, 2N7482, 2N7491, 2N7494, 2N7480, 2N7483, 2N7492, 2N7495.

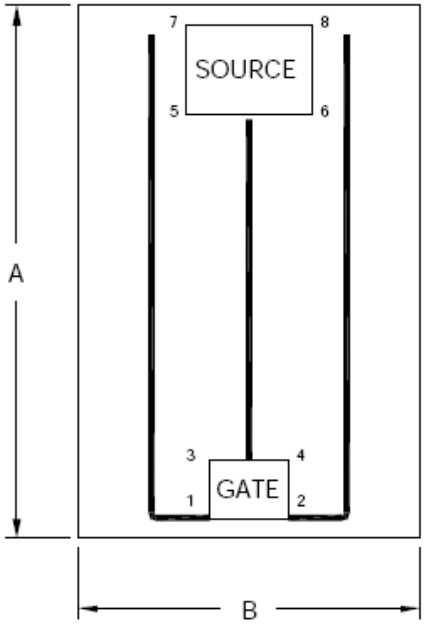


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7481, 2N7484, 2N7493, 2N7496**

| Ltr | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4585.0 | 180.5 | ±200 µm (7.87 mils) |
| B | 2927.0 | 115.2 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 509.9 | 0 | 20.075 | |
| 4 | 685.9 | 509.9 | 27.004 | 20.075 | |
| 5 | -206.1 | 3396.7 | -8.114 | 133.72 | |
| 6 | 891.9 | 3396.7 | 35.114 | 133.72 | |
| 7 | -206.1 | 4153.8 | -8.114 | 163.54 | |
| 8 | 891.9 | 4153.8 | 35.114 | 163.54 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.5 µm (0.22 mils) thick.
3. Back metal: Ti, NiV, Ag, 0.1, 0.45, 0.6 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 290 µm (11.4 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 5. JANHC and JANKC A-version die dimensions, 2N7481, 2N7484, 2N7493, and 2N7496.

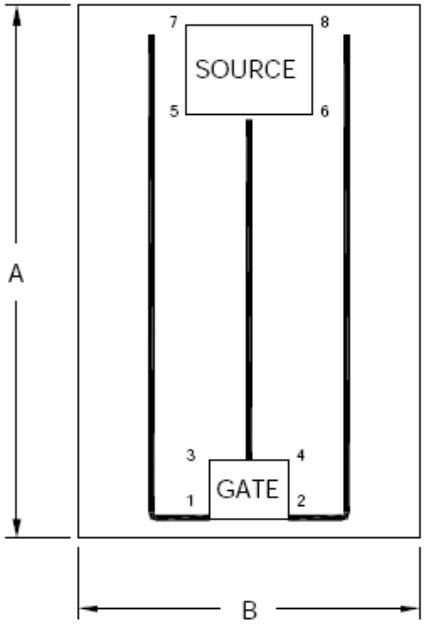


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7545, 2N7547**

| Ltr | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4530.0 | 178.4 | ±200 µm (7.87 mils) |
| B | 2872.0 | 113.1 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 508.0 | 0 | 20.0 | |
| 4 | 685.9 | 508.0 | 27.004 | 20.0 | |
| 5 | -206.1 | 3396.6 | -8.114 | 133.7 | |
| 6 | 871.9 | 3396.6 | 34.327 | 133.7 | |
| 7 | -206.1 | 4153.7 | -8.114 | 163.5 | |
| 8 | 871.9 | 4153.7 | 34.327 | 163.5 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Al-1%Si-0.5%Cu, 5.5 µm (0.22 mils) thick.
3. Back metal: Al/Ti/NiV/Ag; 0.2 µm/0.2 µm/0.45 µm/0.6 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 245 µm (9.6 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 6. JANHC and JANKC A-version die dimensions, 2N7545, and 2N7547.

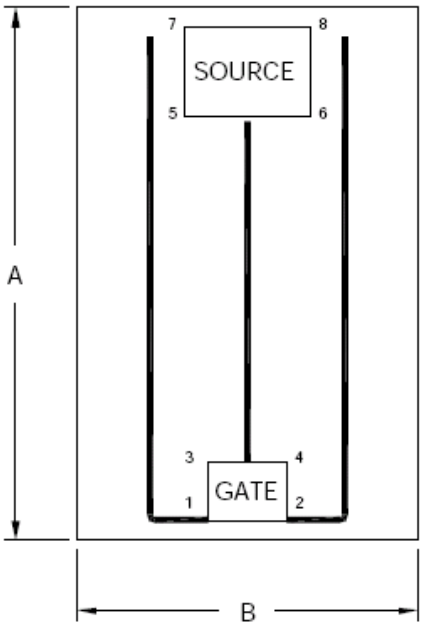


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7485, 2N7488, 2N7497, 2N7500, 2N7486, 2N7489, 2N7498, 2N7501, 2N7555, 2N7561, 2N7490, 2N7502**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4604.3 | 181.3 | ±200 µm (7.87 mils) |
| B | 2946.4 | 116.0 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 508.0 | 0 | 20.000 | |
| 4 | 685.9 | 508.0 | 27.004 | 20.000 | |
| 5 | -206.1 | 3357.3 | -8.114 | 132.18 | |
| 6 | 891.9 | 3357.3 | 35.114 | 132.18 | |
| 7 | -206.1 | 4358.0 | -8.114 | 171.58 | |
| 8 | 891.9 | 4358.0 | 35.114 | 171.58 | |



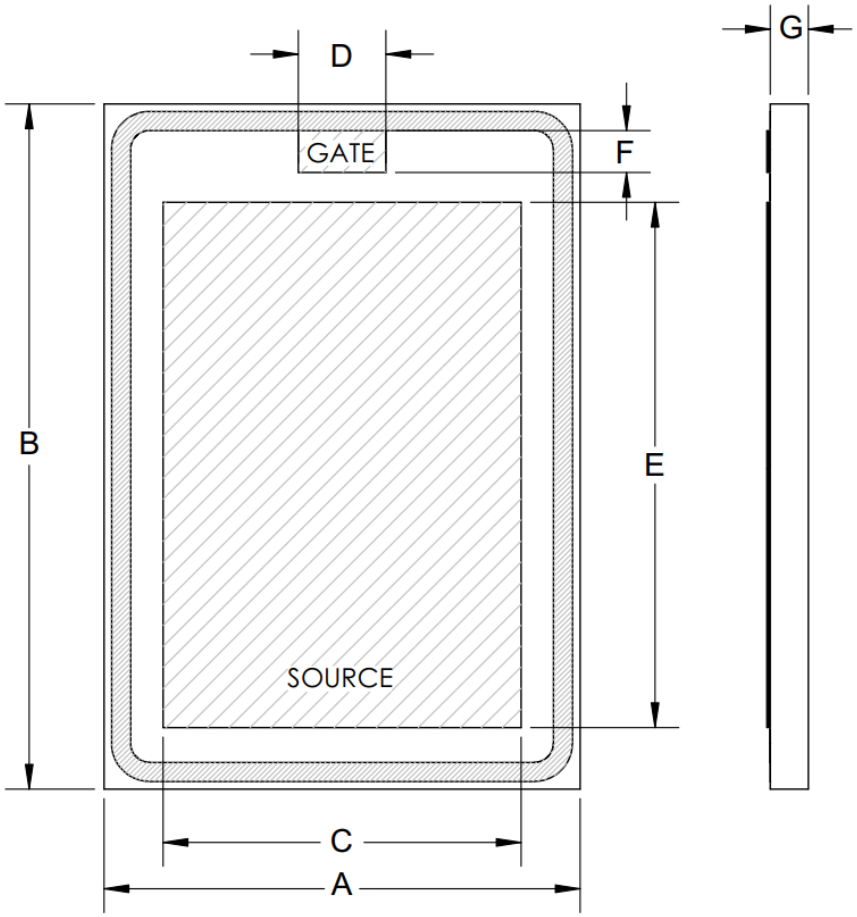

NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.6 µm (0.22 mils) thick.
3. Back metal: Chromium, nickel, Vanadium-silver, 0.1, 0.2, 0.4 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 350 µm (13.77 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 7. JANHC and JANKC A-version die dimensions, 2N7485, 2N7488, 2N7497, 2N7500, 2N7486, 2N7489, 2N7498, 2N7501, 2N7555, 2N7490, 2N7561, and 2N7502.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7491, 2N7492, 2N7493, 2N7494, 2N7495, 2N7496, 2N7479, 2N7480, 2N7481 2N7482, 2N7483, 2N7484, 2N7485, 2N7486, 2N7487, 2N7497, 2N7498, 2N7499, 2N7500, 2N7501, 2N7502, 2N7562, 2N7555, 2N7561**




| Ltr | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| A | 0.120 | 0.130 | 3.05 | 3.30 |
| B | 0.175 | 0.185 | 4.44 | 4.70 |
| C | 0.089 | 0.099 | 2.26 | 2.51 |
| D | 0.020 | 0.026 | 0.51 | 0.66 |
| E | 0.133 | 0.143 | 3.38 | 3.63 |
| F | 0.009 | 0.013 | 0.23 | 0.33 |
| G | 0.012 | 0.016 | 0.30 | 0.41 |

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Die thickness = .014 ±.003 inch (0.36 ±0.07 mm).
3. Back metal: Titanium 1.15KÅ, Nickel 2KÅ, and Silver 8KÅ. Nominal.
4. Top metal: Al 40KÅ. Nominal.
5. Back contact: Drain.

* FIGURE 8. JANHCB and JANKCB (B-version) die dimensions for 2N7491, 2N7492, 2N7493, 2N7494, 2N7495, 2N7496, 2N7479, 2N7480, 2N7481 2N7482, 2N7483, 2N7484, 2N7485, 2N7486, 2N7487, 2N7497, 2N7498, 2N7499, 2N7500, 2N7501, 2N7502, 2N7562, 2N7555, 2N7561.

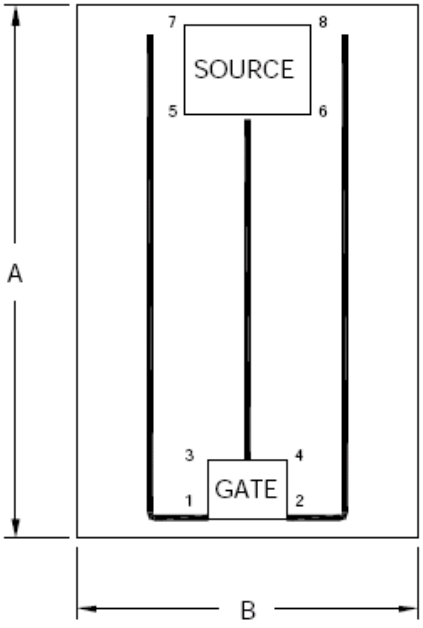


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7546, 2N7548**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4604.3 | 181.3 | ±200 µm (7.87 mils) |
| B | 2946.4 | 116.0 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 508.0 | 0 | 20.000 | |
| 4 | 685.9 | 508.0 | 27.004 | 20.000 | |
| 5 | -206.1 | 3357.3 | -8.114 | 132.18 | |
| 6 | 891.9 | 3357.3 | 35.114 | 132.18 | |
| 7 | -206.1 | 4358.0 | -8.114 | 171.58 | |
| 8 | 891.9 | 4358.0 | 35.114 | 171.58 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.6 µm (0.22 mils) thick.
3. Back metal: Chromium, nickel, Vanadium-silver, 0.1, 0.2, 0.4 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 250 µm (9.8 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 9. JANHC and JANKC A-version die dimensions, 2N7546, and 2N7548.

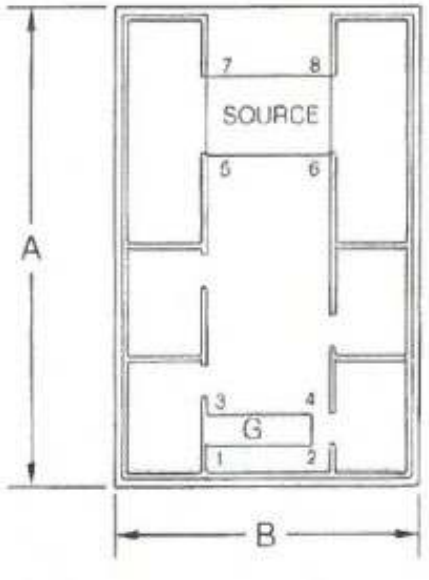


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7519, 2N7520**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 4611 | 181.5 | ±200 µm (7.87 mils) |
| B | 2946.4 | 116.0 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 1041.4 | 0 | 41 | 0 | |
| 3 | 0 | 304.8 | 0 | 12 | |
| 4 | 1041.4 | 304.8 | 41 | 12 | |
| 5 | 34.7 | 2793 | 1.37 | 109.96 | |
| 6 | 1185.7 | 2793 | 46.68 | 109.96 | |
| 7 | 34.7 | 3555 | 1.37 | 139.99 | |
| 8 | 1185.7 | 3555 | 46.68 | 139.99 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.6 (0.22 mils) thick.
3. Back metal: Aluminum, Titanium, nickel-Vanadium, silver, 0.1, 0.1, 0.4, 0.6 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 250 µm (10 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate (G)/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

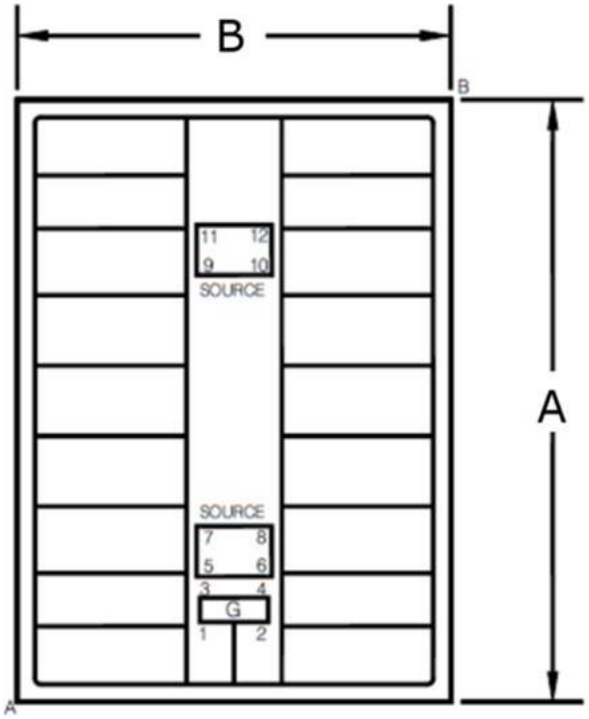
FIGURE 10. JANHC and JANKC A-version die dimensions, 2N7519 and 2N7520.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7523, 2N7524**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 9144 | 360.0 | ±200 µm (7.87 mils) |
| B | 6604 | 260.0 | |




| Key | Pad | Micrometers | | mils | | TCL | Tolerance |
|---|---|---|---|---|---|---|---|
| | | X | Y | X | Y | | |
| 1 | Gate | 0 | 0 | 0 | 0 | +/- 5 µm | 5 µm (0.2 mils) |
| 2 | | 1041.40 | 0 | 41.00 | 0 | +/- 5 µm | |
| 3 | | 0 | 368 | 0 | 14.50 | +/- 5 µm | |
| 4 | | 1041.40 | 368 | 41.00 | 14.50 | +/- 5 µm | |
| 5 | Source A | -50.80 | 692.15 | -2.00 | 27.25 | +/- 5 µm | |
| 6 | | 1092.20 | 692.15 | 43.00 | 27.25 | +/- 5 µm | |
| 7 | | -50.80 | 1454.15 | -2.00 | 57.25 | +/- 5 µm | |
| 8 | | 1092.20 | 1454.15 | 43.00 | 57.25 | +/- 5 µm | |
| 9 | Source B | -50.80 | 5264.15 | -2.00 | 207.25 | +/- 5 µm | |
| 10 | | 1092.20 | 5264.15 | 43.00 | 207.25 | +/- 5 µm | |
| 11 | | -50.80 | 6026.15 | -2.00 | 237.25 | +/- 5 µm | |
| 12 | | 1092.20 | 6026.15 | 43.00 | 237.25 | +/- 5 µm | |

NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 8 µm (0.31 mils) thick.
3. Back metal: Titanium, nickel, silver, 0.1, 0.2, 0.25 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 250 µm (9.8 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 12 refer to adjacent gate (G)/source pad corners. Key 2 through 12 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 11. JANHC and JANKC A-version die dimensions, 2N7523 and 2N7524.

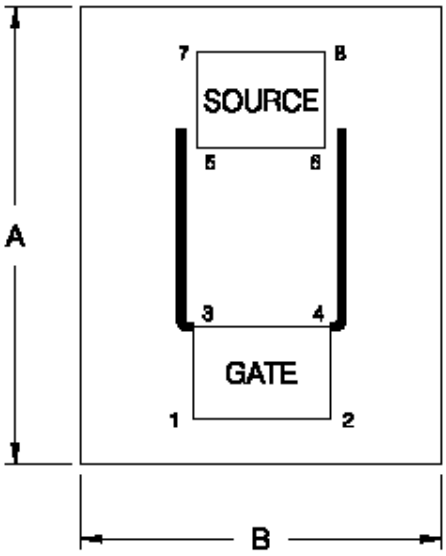


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7503, N or P channel, 2N7521, 2N7525**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 2542.9 | 100.1 | ±200 µm (7.87 mils) |
| B | 2010.0 | 79.1 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 502.0 | 0 | 19.764 | |
| 4 | 685.9 | 502.0 | 27.004 | 19.764 | |
| 5 | -10.0 | 1560.5 | -0.394 | 61.437 | |
| 6 | 695.8 | 1560.5 | 27.394 | 61.437 | |
| 7 | -10.0 | 2099.9 | -0.394 | 82.673 | |
| 8 | 695.8 | 2099.9 | 27.394 | 82.673 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.6 µm (0.22 mils) thick.
3. Back metal: Chromium, Nickel, Vanadium-silver, 0.1, 0.2, 0.5 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 305 µm (12 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 12. JANHC and JANKC A-version die dimensions, 2N7503, 2N7521, 2N7525.

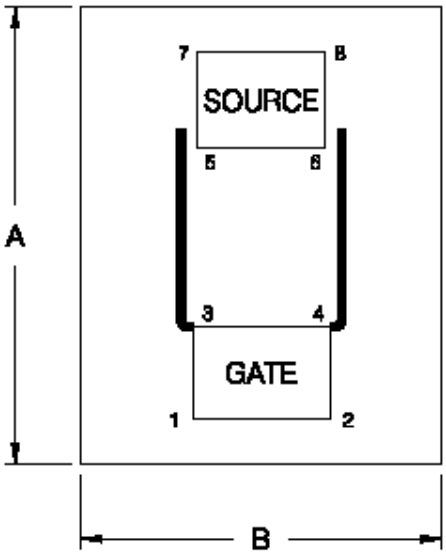


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7506, 2N7522, 2N7526**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 2542.9 | 100.1 | ±200 µm (7.87 mils) |
| B | 2010.0 | 79.1 | |

| Key | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|
| | Micrometers | | Mils | | |
| | X | Y | X | Y | |
| 1 | 0 | 0 | 0 | 0 | ±5 µm (0.2 mils) |
| 2 | 685.9 | 0 | 27.004 | 0 | |
| 3 | 0 | 502.0 | 0 | 19.764 | |
| 4 | 685.9 | 502.0 | 27.004 | 19.764 | |
| 5 | -10.0 | 1560.5 | -0.394 | 61.437 | |
| 6 | 695.8 | 1560.5 | 27.394 | 61.437 | |
| 7 | -10.0 | 2099.9 | -0.394 | 82.673 | |
| 8 | 695.8 | 2099.9 | 27.394 | 82.673 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Aluminum, 5.6 µm (0.22 mils) thick.
3. Back metal: Chromium, Nickel, Vanadium-silver, 0.1, 0.2, 0.5 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 250 µm (9.8 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

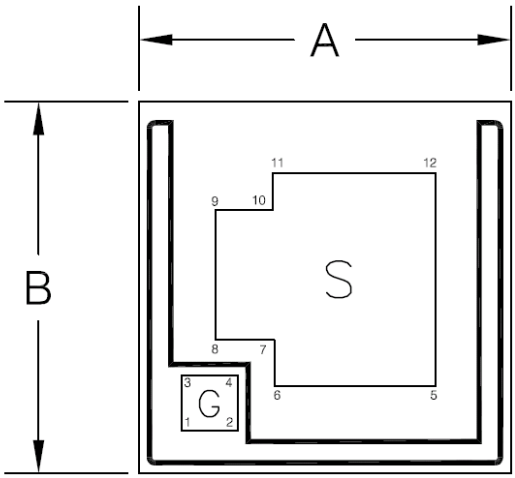
FIGURE 13. JANHC and JANKC A-version die dimensions, 2N7506, 2N7522, and 2N7526.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7616**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 1016.00 | 40.0 | ±200 µm (7.87 mils) |
| B | 1014.80 | 39.95 | |




| Key | Pad | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|---|
| | | Micrometers | | Mils | | |
| | | X | Y | X | Y | |
| 1 | Gate | 0.00 | 0.00 | 0.00 | 0.00 | ±5 µm (0.2 mils) |
| 2 | | 152.40 | 0.00 | 6.00 | 0.00 | |
| 3 | | 0.00 | 152.20 | 0.00 | 5.99 | |
| 4 | | 152.40 | 152.20 | 6.00 | 5.99 | |
| 5 | Source | 678.10 | 106.50 | 26.70 | 4.19 | |
| 6 | | 238.20 | 106.5 | 9.38 | 4.19 | |
| 7 | | 238.20 | 233.30 | 9.38 | 9.19 | |
| 8 | | 81.50 | 233.30 | 3.21 | 9.19 | |
| 9 | | 81.50 | 590.70 | 3.21 | 23.26 | |
| 10 | | 238.20 | 590.70 | 9.38 | 23.26 | |
| 11 | | 238.20 | 690.70 | 9.38 | 27.19 | |
| 12 | | 678.10 | 690.70 | 26.70 | 27.19 | |

NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Al-1%Si-0.5%Cu; 5.7 µm thick respectively.
3. Back metal: Ti/NiV/Ag; 0.1 µm / 0.45 µm / 0.6 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 240 µm (9.4 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 12 refer to adjacent gate/source pad corners. Key 2 through 12 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 14. JANHC and JANKC A-version die dimensions, 2N7616.

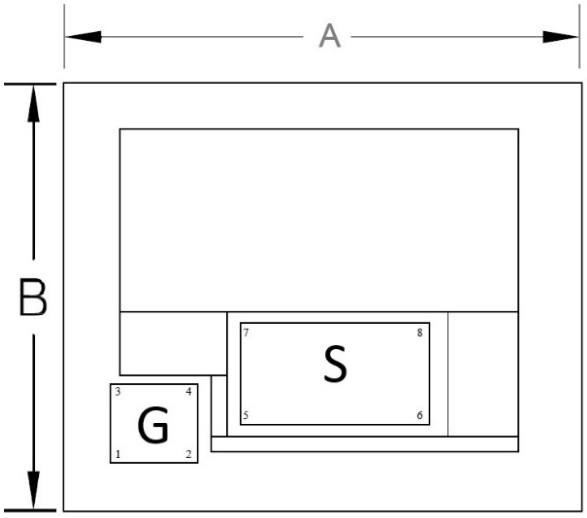


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7626**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| A | 1016.00 | 40.0 | ±200 µm (7.87 mils) |
| B | 1014.80 | 39.95 | |

| Key | Pad | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|---|
| | | Micrometers | | Mils | | |
| | | X | Y | X | Y | |
| 1 | Gate | 00.0 | 0.00 | 0.00 | 0.00 | ±5 µm (0.2 mils) |
| 2 | | 152.40 | 0.00 | 6.00 | 0.00 | |
| 3 | | 0.00 | 152.20 | 0.00 | 5.99 | |
| 4 | | 152.40 | 152.20 | 6.00 | 5.99 | |
| 5 | Source | 236.50 | 86.60 | 9.31 | 3.41 | |
| 6 | | 617.50 | 86.60 | 24.31 | 3.41 | |
| 7 | | 236.50 | 300.20 | 9.31 | 11.82 | |
| 8 | | 617.50 | 300.20 | 24.31 | 11.82 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: Al-1%Si-0.5%Cu; 5.7 µm thick.
3. Back metal: Al/Ti/NiV/Ag; 0.1 µm/0.1 µm/0.45 µm/0.6 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 240 µm (9.4 mils).
7. For sawn die, outline dimensions (A and B) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 15. JANHC and JANKC A-version die dimensions, 2N7626.

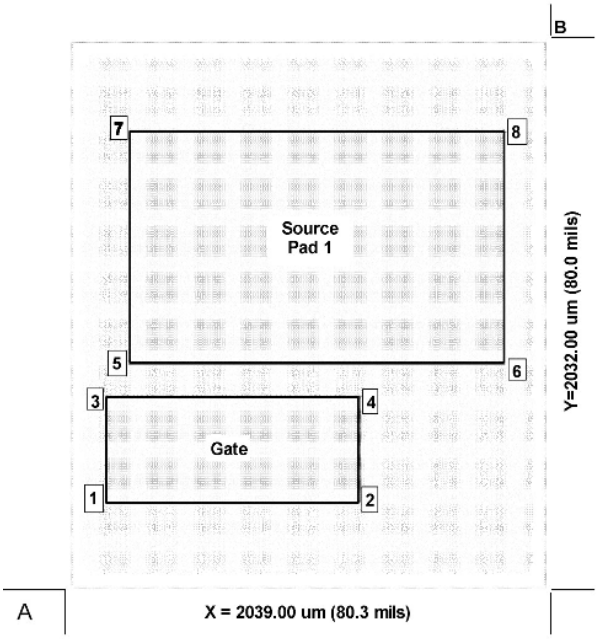


Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

**2N7656**

| Ltr. | Dimensions | | Tolerance |
|---|---|---|---|
| | Micrometers | Mils | |
| X | 2039.0 | 80.3 | ±200 µm (7.87 mils) |
| Y | 2032.0 | 80.0 | |

| Key | Pad | Die bond pad coordinates | | | | Tolerance |
|---|---|---|---|---|---|---|
| | | Micrometers | | Mils | | |
| | | X | Y | X | Y | |
| 1 | Gate | 00.0 | 0.00 | 0.0 | 0.0 | ±5 µm (0.2 mils) |
| 2 | | 1042.0 | 0.00 | 41.0 | 0.0 | |
| 3 | | 0.0 | 369.0 | 0.0 | 14.5 | |
| 4 | | 1042.0 | 369.0 | 6.0 | 14.5 | |
| 5 | Source | 109.3 | 496.0 | 4.3 | 19.5 | |
| 6 | | 1633.3 | 496.0 | 64.3 | 19.5 | |
| 7 | | 109.3 | 1308.8 | 4.3 | 51.5 | |
| 8 | | 1633.3 | 1308.8 | 64.3 | 51.5 | |
| Ref Point | A | -185.3 | .355.6 | -7.3.0 | -14.0 | ±200 µm (7.87 mils) |
| | B | 1853.7 | 1676.4 | 73.0 | 66.0 | |




NOTES:
1. Dimensions are in micrometers. Mils are given for general information only.
2. Top metal: 5.5 µm thick.
3. Back metal: Titanium, nickel, silver; 0.1, 0.45, 0.6 µm thick, respectively.
4. All dimensions are valid for all radiation hardness levels specified.
5. Backside metal is the drain connection.
6. Die thickness: 189 µm (7.4 mils).
7. For sawn die, outline dimensions (X and Y) will be reduced by 25 µm (0.98 mils), due to saw kerf.
8. Die bond pad coordinates are provided for use in automated bonding equipment. Key locations 1 through 8 refer to adjacent gate/source pad corners. Key 2 through 8 are relative to Key 1.
9. See 6.2 and 6.5 for ordering information.

FIGURE 16. JANHC and JANKC A-version die dimensions, 2N7656.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 through 15.

3.5 Marking. The individual die are not required to be marked. The die container (waffle pack) shall be marked in accordance with MIL-PRF-19500.

3.6 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Avoid use of plastic, rubber, or silk in MOS areas.

d. Maintain relative humidity above 50 percent if practical.

e. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any pad.

f. Gate must be terminated to source, R ≤ 100 kΩ, whenever bias voltage is to be applied drain to source.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in the applicable specification sheet listed in table I herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (element evaluation) (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

4.2 Qualification inspection (element evaluation). Qualification inspection (element evaluation) shall be in accordance with appendix G of MIL-PRF-19500 and the applicable specification sheet from table I herein.

4.3 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with appendix G of MIL-PRF-19500 and the applicable specification sheet from table I herein, except test current shall not exceed 20 A. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTXV requirements.

4.4 Conformance inspection (group D). Conformance inspection (group D) shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and the applicable performance specification sheet from table I herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

* TABLE I. Applicable specification sheets.

| Type (same exact die in:) | Specification sheet and common part number | Reference data | | | Figure |
|---|---|---|---|---|---|
| | | Voltage (V dc) | Channel | Die size | |
| 2N7467 or 2N7478 | MIL-PRF-19500/697, 2N7478 | 30 | N | 6 | 1 |
| 2N7467 or 2N7478 | MIL-PRF-19500/683, 2N7467 | 30 | N | 6 | 1 |
| 2N7468 or 2N7470 | MIL-PRF-19500/698, 2N7470 | 60 | N | 6 | 1 |
| 2N7469 or 2N7471 | MIL-PRF-19500/698, 2N7471 | 100 | N | 6 | 2 |
| 2N7472 or 2N7475 | MIL-PRF-19500/685, 2N7475 | 130 | N | 6 | 2 |
| 2N7473 or 2N7476 | MIL-PRF-19500/685, 2N7476 | 200 | N | 6 | 2 |
| 2N7474 or 2N7477 | MIL-PRF-19500/685, 2N7477 | 250 | N | 6 | 2 |
| 2N7479 or 2N7482 or 2N7491 or 2N7494 | MIL-PRF-19500/702, 2N7482 | 30 | N | 3 | 4, 8 |
| 2N7480 or 2N7483 or 2N7492 or 2N7495 | MIL-PRF-19500/702, 2N7483 | 60 | N | 3 | 4, 8 |
| 2N7481 or 2N7484 or 2N7493 or 2N7496 | MIL-PRF-19500/702, 2N7484 | 100 | N | 3 | 5, 8 |
| 2N7485 or 2N7488 or 2N7497 or 2N7500 | MIL-PRF-19500/705, 2N7488 | 130 | N | 3 | 7, 8 |
| 2N7486 or 2N7489 or 2N7498 or 2N7501 | MIL-PRF-19500/705, 2N7489 | 200 | N | 3 | 7, 8 |
| 2N7555 or 2N7487 or 2N7490 or 2N7499 or 2N7561 or 2N7502 or 2N7562 | MIL-PRF-19500/705, 2N7490 | 250 | N | 3 | 7, 8 |
| 2N7503 or 2N7521 or 2N7525 | MIL-PRF-19500/740, 2N7525 | 100 | N | 1 | 12 |
| 2N7523 | MIL-PRF-19500/733, 2N7523 | -30 | P | 6 | 11 |
| 2N7524 | MIL-PRF-19500/733, 2N7524 | -60 | P | 6 | 11 |
| 2N7550 | MIL-PRF-19500/713, 2N7550 | -100 | P | 6 | 3 |
| 2N7549 | MIL-PRF-19500/713, 2N7549 | -200 | P | 6 | 3 |
| 2N7519 | MIL-PRF-19500/732, 2N7519 | -30 | P | 3 | 10 |
| 2N7520 | MIL-PRF-19500/732, 2N7520 | -60 | P | 3 | 10 |
| 2N7545 or 2N7547 | MIL-PRF-19500/712, 2N7547 | -100 | P | 3 | 6 |
| 2N7546 or 2N7548 | MIL-PRF-19500/712, 2N7548 | -200 | P | 3 | 9 |
| 2N7506 or 2N7522 or 2N7526 | MIL-PRF-19500/740, 2N7526 | -100 | P | 1 | 13 |
| 2N7616 | MIL-PRF-19500/744, 2N7616 | 60 | N | 7 | 14 |
| 2N7626 | MIL-PRF-19500/745, 2N7626 | -60 | P | 7 | 15 |
| 2N7656 | MIL-PRF-19500/790, 2N7656 | 100 | N | 1 | 16 |



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

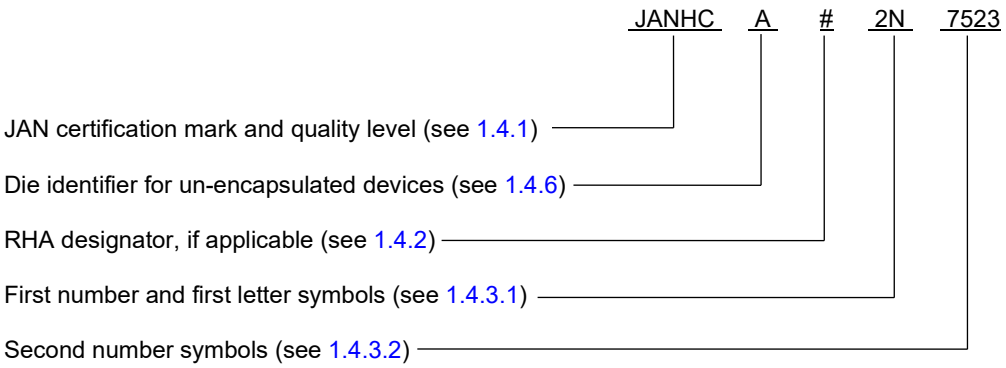
c. The complete PIN, see 1.4 and 6.5.

d. Specify the JANHC or JANKC letter version (see 6.5 and figures 1 through 15 herein).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QPDSIS-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Weapons Support, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil/.

6.4 PIN construction example.

6.4.1 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.






Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

* 6.5 List of PINs for un-encapsulated devices. The following is a list of possible PINs available on this specification sheet. The qualified die suppliers with the applicable letter version (example, JANHCA2N7261) will be identified on the QML-19500.

| JANC ordering information | | |
|---|---|---|
| For die in type | Manufacturer CAGE 69210 | Manufacturer CAGE 52GC4 |
| | Common JANC die part numbers for ordering 1/ | Common JANC die part numbers for ordering 1/ |
| 2N7467 or 2N7478 | JANHCAR2N7478, JANKCAR2N7478, JANHCAF2N7478, JANKCAF2N7478 JANHCAG2N7478, JANKCAG2N7478, JANHCAH2N7478, JANKCAH2N7478 | |
| 2N7468 or 2N7470 | JANHCAR2N72470, JANKCAR2N7470, JANHCAF2N7470, JANKCAF2N7470 JANHCAG2N72470, JANKCAG2N7470, JANHCAH2N7470, JANKCAH2N7470 | |
| 2N7469 or 2N7471 | JANHCAR2N7471, JANKCAR2N7471, JANHCAF2N7471, JANKCAF2N7471 JANHCAG2N7471, JANKCAG2N7471, JANHCAH2N7471, JANKCAH2N7471 | |
| 2N7472 or 2N7475 | JANHCAR2N7475, JANKCAR2N7475, JANHCAF2N7475, JANKCAF2N7475 | |
| 2N7473 or 2N7476 | JANHCAR2N7476, JANKCAR2N7476, JANHCAF2N7476, JANKCAF2N7476 | |
| 2N7474 or 2N7477 | JANHCAR2N7477, JANKCAR2N7477, JANHCAF2N7477, JANKCAF2N7477 | |
| * 2N7479 or 2N7482 or 2N7491 or 2N7494 | JANHCAR2N7482, JANKCAR2N7482, JANHCAF2N7482, JANKCAF2N7482 JANHCAG2N7482, JANKCAG2N7482, JANHCAH2N7482, JANKCAH2N7482 | JANHCBR2N7482, JANKCBR2N7482, JANHCBF2N7482, JANKCBF2N7482 JANHCBG2N7482, JANKCBG2N7482, JANHCBH2N7482, JANKCBH2N7482 |
| * 2N7480 or 2N7483 or 2N7492 or 2N7495 | JANHCAR2N7483, JANKCAR2N7483, JANHCAF2N7483, JANKCAF2N7483 JANHCAG2N7483, JANKCAG2N7483, JANHCAH2N7483, JANKCAH2N7483 | JANHCBR2N7483, JANKCBR2N7483, JANHCBF2N7483, JANKCBF2N7483 JANHCBG2N7483, JANKCBG2N7483, JANHCBH2N7483, JANKCBH2N7483 |
| * 2N7481 or 2N7484 or 2N7493 or 2N7496 | JANHCAR2N7484, JANKCAR2N7484, JANHCAF2N7484, JANKCAF2N7484 JANHCAG2N7484, JANKCAG2N7484, JANHCAH2N7484, JANKCAH2N7484 | JANHCBR2N7484, JANKCBR2N7484, JANHCBF2N7484, JANKCBF2N7484 JANHCBG2N7484, JANKCBG2N7484, JANHCBH2N7484, JANKCBH2N7484 |
| * 2N7485 or 2N7488 or 2N7497 or 2N7500 | JANHCAR2N7488, JANKCAR2N7488, JANHCAF2N7488, JANKCAF2N7488 | JANHCBR2N7488, JANKCBR2N7488, JANHCBF2N7488, JANKCBF2N7488 |
| * 2N7486 or 2N7489 or 2N7498 or 2N7501 | JANHCAR2N7489, JANKCAR2N7489, JANHCAF2N7489, JANKCAF2N7489 | JANHCBR2N7489, JANKCBR2N7489, JANHCBF2N7489, JANKCBF2N7489 |
| * 2N7555 or 2N7487 or 2N7490 or 2N7499 or 2N7561 or 2N7502 or 2N7562 | JANHCAR2N7490, JANKCAR2N7490, JANHCAF2N7490, JANKCAF2N7490 | JANHCBR2N7490, JANKCBR2N7490, JANHCBF2N7490, JANKCBF2N7490 |
| 2N7503 or 2N7521 or 2N7525 | JANHCAR2N7525, JANKCAR2N7525, JANHCAF2N7525, JANKCAF2N7525 JANHCAG2N7525, JANKCAG2N7525, JANHCAH2N7525, JANKCAH2N7525 | |
| 2N7523 | JANHCAR2N7523, JANKCAR2N7523, JANHCAF2N7523, JANKCAF2N7523 | |

See footnotes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

| JANC ordering information -Continued. | | |
|---|---|---|
| For die in type | Manufacturer CAGE 69210 | Manufacturer CAGE 52GC4 |
| | Common JANC die part numbers for ordering 1/ | Common JANC die part numbers for ordering 1/ |
| 2N7524 | JANHCAR2N7524, JANKCAR2N7524, JANHCAF2N7524, JANKCAF2N7524 | |
| 2N7550 | JANHCAR2N7550, JANKCAR2N7550, JANHCAF2N7550, JANKCAF2N7550 | |
| 2N7549 | JANHCAR2N7549, JANKCAR2N7549, JANHCAF2N7549, JANKCAF2N7549 | |
| 2N7519 | JANHCAR2N7519, JANKCAR2N7519, JANHCAF2N7519, JANKCAF2N7519 | |
| 2N7520 | JANHCAR2N7520, JANKCAR2N7520, JANHCAF2N7520, JANKCAF2N7520 | |
| 2N7545 or 2N7547 | JANHCAR2N7547, JANKCAR2N7547, JANHCAF2N7547, JANKCAF2N7547 | |
| 2N7546 or 2N7548 | JANHCAR2N7548, JANKCAR2N7548, JANHCAF2N7548, JANKCAF2N7548 | |
| 2N7506 or 2N7522 or 2N7526 | JANHCAR2N7526, JANKCAR2N7526, JANHCAF2N7526, JANKCAF2N7526 | |
| 2N7616 | JANHCAR2N7616, JANHCAF2N7616, JANKCAR2N7616, JANKCAF2N7616 | |
| 2N7626 | JANHCAR2N7626, JANHCAF2N7626, JANKCAR2N7626, JANKCAF2N7626 | |
| 2N7656 | JANHCAR2N7656, JANHCAF2N7656, JANKCAR2N7656, JANKCAF2N7656 | |

1/ The PIN is also available without a RHA designator.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

6.6 Cross reference list. The following information shows the generic P/N and its' associated military P/N (without the JAN or RHA prefix). Multiple military part number indicates that the same die type is used on more than one specification sheet.

| For any military die-packaged basic P/N | Common military die P/N (1) | Generic die P/N (2) |
|---|---|---|
| 2N7467 or 2N7478 | JANHCAX or JANKCAX2N7478 | IRHC5xZ60 |
| 2N7468 or 2N7470 | JANHCAX or JANKCAX2N7470 | IRHC5x064 |
| 2N7469 or 2N7471 | JANHCAX or JANKCAX2N7471 | IRHC5x160 |
| 2N7472 or 2N7475 | JANHCAX or JANKCAX2N7475 | IRHC5x163SE |
| 2N7473 or 2N7476 | JANHCAX or JANKCAX2N7476 | IRHC5x260SE |
| 2N7474 or 2N7477 | JANHCAX or JANKCAX2N7477 | IRHC5x264SE |
| 2N7479 or 2N7482 or 2N7491 or 2N7494 | JANHCAX or JANKCAX2N7482 | IRHC5xZ30 |
| 2N7480 or 2N7483 or 2N7492 or 2N7495 | JANHCAX or JANKCAX2N7483 | IRHC5x034 |
| 2N7481 or 2N7484 or 2N7493 or 2N7496 | JANHCAX or JANKCAX2N7484 | IRHC5x130 |
| 2N7485 or 2N7488 or 2N7497 or 2N7500 | JANHCAX or JANKCAX2N7488 | IRHC5x133SE |
| 2N7486 or 2N7489 or 2N7498 or 2N7501 | JANHCAX or JANKCAX2N7489 | IRHC5x230SE |
| 2N7555 or 2N7490 or 2N7561 or 2N7502 | JANHCAX or JANKCAX2N7490 | IRHC5x234SE |
| 2N7521 or 2N7525 or 2N7503 | JANHCAX or JANKCAX2N7525 | IRHC5x110 |
| 2N7523 | JANHCAX or JANKCAX2N7523 | IRHC59xZ60 |
| 2N7524 | JANHCAX or JANKCAX2N7524 | IRHC59x064 |
| 2N7550 | JANHCAX or JANKCAX2N7550 | IRHC59x160 |
| 2N7549 | JANHCAX or JANKCAX2N7549 | IRHC59x260 |
| 2N7519 | JANHCAX or JANKCAX2N7519 | IRHC59xZ30 |
| 2N7520 | JANHCAX or JANKCAX2N7520 | IRHC59x034 |
| 2N7545 or 2N7547 | JANHCAX or JANKCAX2N7547 | IRHC59x130 |
| 2N7546 or 2N7548 | JANHCAX or JANKCAX2N7548 | IRHC59x230 |
| 2N7522 or 2N7526 or 2N7506 | JANHCAX or JANKCAX2N7526 | IRHC59x110 |
| 2N7616 | JANHCAX or JANKCAX2N7616 | IRHL7770Z4 |
| 2N7626 | JANHCAX or JANKCAX2N7626 | IRHL7970Z4 |
| 2N7656 | JANHCAX or JANKCAX2N7656 | IRHC9A7110 |

(1) These common die part numbers should be used for the same exact die used in their respective die-packaged part numbers. Replace X with letter indicating qualified rad-hardness (RHA designator) as follows:

R = 100 K Rad (Si) (equivalent to manufacturer designator 7).
F = 300 K Rad (Si) (equivalent to manufacturer designator 3).
G = 500 K Rad (Si) (equivalent to manufacturer designator 4).
H = 1000 K Rad (Si) (equivalent to manufacturer designator 8).

(2) Replace x with number indicating qualified rad-hardness as follows:

7 = 100 K Rad (Si) (equivalent to military RHA designator R).
3 = 300 K Rad (Si) (equivalent to military RHA designator F).
4 = 500 K Rad (Si) (equivalent to military RHA designator G).
8 = 1000 K Rad (Si) (equivalent to military RHA designator H).



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.

6.7 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Weapons Support, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil.

6.8 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA
DLA - CC

Review activity:
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2026-043)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



Source: https://assist.dla.mil -- Downloaded: 2026-10-19T05:23Z
Check the source to verify that this is the current version before use.